DATA PIPELINES & DASHBOARDS · DATA ANALYTICS AND DATA ENGINEERING

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Data analytics, data engineering and data visualization: pipelines on AWS, Snowflake and BigQuery, and dashboards in Looker Studio and Power BI

github.com/debiegalleros debiegalleros.github.io/pipelines-and-dashboards

I work across the whole path of data. I take data from where it starts, such as ad platforms, website tracking and seller reports, and move it into a database or warehouse with scheduled jobs and SQL. Then I build the dashboards and reports that teams use to follow their campaigns, websites and sales.

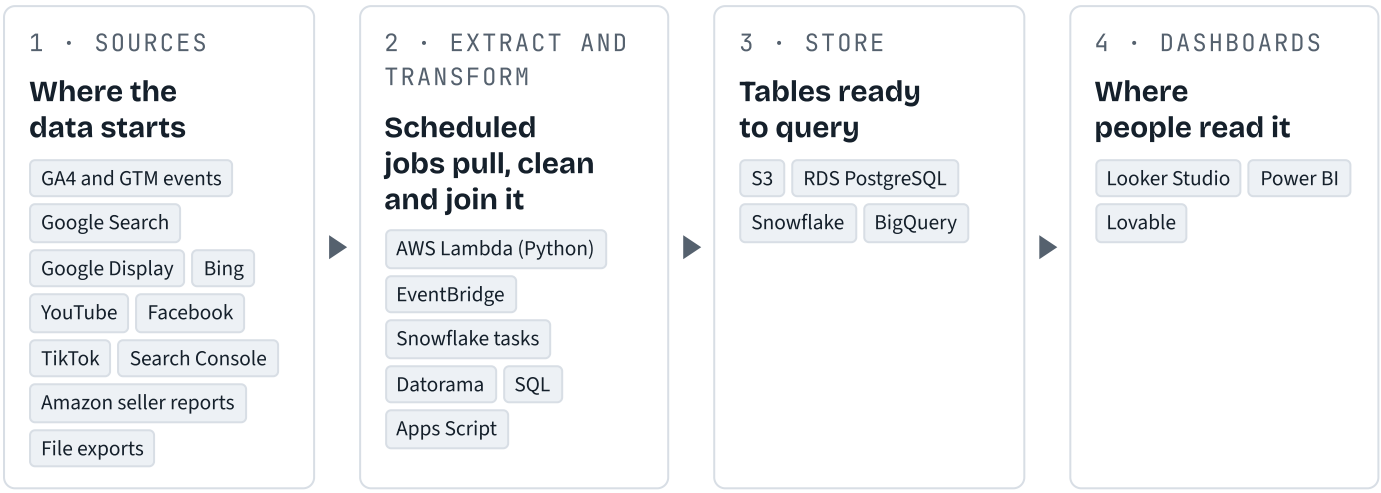
This page shows how the data flows, nine projects, recreations of twelve of my dashboards, a sample quarterly report, my GA4 tracking work, and short code examples. Client names and real numbers are left out, so every chart here uses sample data.

WHAT I DO, AND THE TOOLS I USE

TASK	AWS	SNOWFLAKE	BIGQUERY	SQL	PYTHON	GA4 & GTM	LOOKER STUDIO, POWER BI	AI (LLM)	CASE STUDIES
Build data pipelines Scheduled jobs that load reports as they arrive, copy only new rows, and never publish an incomplete month.	●	●	●	●	●				1, 2, 7
Model and check data SQL that joins sources at the right grain, with checks that row counts and totals match the source.	●	●	●	●					1, 2, 7
Track websites GA4 events and custom dimensions, planned with the developer who sets up the tags.						●			3, 4, 5, 6
Build dashboards Summary first, then trends, then detail, with every metric defined on the page.			●	●		●	●		3 – 8
Report insights Monthly and quarterly reports and decks that start with the answer.						●	●	●	4, 6, 9
Analyse text with AI A large language model and one fixed prompt to sort thousands of comments into themes.								●	9

How the data flows

The path most of my projects follow, from where the data starts to the dashboard people read. Each project uses some of these tools, not all of them.



Checks along the way: every required file is there before a job runs, no row is loaded twice, and totals match across every output.

Case studies

DATA ENGINEERING · AWS, POSTGRESQL

1 Seller reports, from upload to weekly review

PROBLEM	Weekly business reviews needed sales, advertising, fees and inventory for each product in one table. The team built it by hand, copying each Seller Central CSV into Excel, and every report had its own layout.
DATA	10 report types, including sales, sponsored ads, transactions, orders and two inventory reports.
WHAT I BUILT	A Lambda function that runs when a file lands in S3. It finds the report type from the file name, skips extra header rows, cleans the columns, loads the rows into PostgreSQL on RDS, and moves the file to a processed folder. A stored procedure adds only rows the archive doesn't have yet, and a SQL view joins every report by brand, day and product. The database sits in a private network, so I wrote the team's guide to reach it through an SSH tunnel in VS Code, with a query that checks new uploads arrive.
OUTCOME	**No more copying CSVs into Excel: one table per brand, day and product with ad spend, sales, conversion rate, inventory cover and fees.** New reports show up as soon as they're uploaded, every brand is reviewed from the same numbers, and fees on orders with several products count once.

DATA ENGINEERING · AWS, SNOWFLAKE

2 Monthly brand reports on a schedule

PROBLEM Each brand needed a monthly report by product, with weekly trends and a comparison with the month before. Data shared with us in Snowflake also had to reach the reporting database.

DATA Up to 16 Amazon reports per brand each month, saved in S3 by report type, brand and month: sales and traffic, sponsored ads, orders, transactions, listings, inventory, search performance and SKU economics. Daily, product and search query performance shared in Snowflake.

WHAT I BUILT A monthly Lambda job for each brand, started by its own EventBridge rule instead of by every file upload. It writes nothing if a required report is missing and skips months already done. It writes five files, including a summary, a by-product file, a weekly series and a calendar of Sunday-to-Saturday retail weeks, each with a timestamped backup. In Snowflake, scheduled tasks refresh six mirror tables, and a sync job copies only new rows into PostgreSQL.

OUTCOME **Each brand's monthly report builds itself on a schedule, with weekly, by-product and summary totals that match exactly.** A month with a missing report is held back instead of showing wrong totals, and the mirror tables match their source row for row. I tested in a separate bucket first and watched each run in CloudWatch.

DASHBOARDS · GA4, LOOKER STUDIO DASHBOARD H DASHBOARD I

3 One dashboard for websites in seven countries

PROBLEM A brand ran websites in seven countries, plus a site shoppers reach by scanning the pack. It needed a global view and the same view for each country.

DATA GA4 for each country site: channels, pages, scroll depth, CTA clicks and purchases. Scan data for the pack site.

WHAT I BUILT 16 views, global and per country, for the websites and the scan site. Each website view follows five steps: Catch, Connect, Convert, Webshop and Continue. The scan views show weekly scans, scans by city tier, scan errors and accuracy, and a product breakdown. A glossary page defines every metric.

OUTCOME **All seven markets in one dashboard, on the same metrics.** Teams can compare markets side by side, follow each one through the same five steps, and check what any metric means in the glossary.

WEB ANALYTICS · GA4, GTM, LOOKER STUDIO DASHBOARD A DASHBOARD B DASHBOARD C

4 Measuring a visitor journey, stage by stage

PROBLEM A health-education website wanted to see how far visitors moved, from first interest to taking action, not just how many came.

DATA 29 GA4 events tracked through GTM, such as polls, guide downloads, video plays and finder searches, by channel and device.

WHAT I BUILT I grouped the events into six journey stages and their key actions. A monthly Looker Studio dashboard shows each stage's rate and sessions against the period before, plus traffic by channel and device. Each quarter I wrote a highlights deck with an executive summary, including how much the cookie consent banner limited what GA4 could measure.

OUTCOME **The team can see how far visitors get, not only how many arrive.** Each stage's rate shows where visitors drop off, and the quarterly deck turned the numbers into next steps for the site. The consent check showed how much traffic GA4 couldn't see, so the numbers are read with that in mind.

EVENT TRACKING · GTM, GA4, LOOKER STUDIO DASHBOARD E DASHBOARD F

5 Pop-up survey and call-to-action tracking

PROBLEM — A brand showed a survey pop-up across its site and three calls to action on two campaign sites. It needed to know where people saw them, where they responded, and which traffic brought responses.

DATA — GA4 events for pop-up views, form submits and CTA clicks, with source, medium, device, page and product. Visits from a messaging app's in-app browser were split from the rest.

WHAT I BUILT — Two Looker Studio dashboards with filters for device, source, page type and product: submissions by day and hour, submission rate by page type, CTA click-through over time, and tables by CTA, device, source and channel. Submit tracking started seven weeks after pop-up tracking, so the submission rate only counts from that date, and the dashboard says so.

OUTCOME — **The team can see where the survey and calls to action work, by page, source and device.** Pages are judged by the responses they bring, not just views, and messaging-app visitors can be compared with everyone else. The submission rate stays honest because it leaves out the weeks before submits were tracked.

SEO ANALYTICS · GA4, SEARCH CONSOLE DASHBOARD G

6 A monthly SEO report from two sources

PROBLEM — The team wanted one monthly view of organic search: which pages held up, which searches brought people in, and the trend over a year.

DATA — GA4 sessions, bounce rate and time on site by landing page. Search Console clicks, impressions, click-through rate and position by query and page, with queries in two languages.

WHAT I BUILT — Page groups such as home, guides, product pages and sign-up, each with month-on-month change. Query groups that split branded from generic searches. 12-month trend charts for each page group, and a table of visits referred by AI assistants such as ChatGPT.

OUTCOME — **One monthly view of organic search, from both tools.** Branded and generic searches are kept apart, each page group shows what grew or fell, and visits from AI assistants are tracked as their own source.

DIGITAL CAMPAIGNS · DATORAMA, BIGQUERY, LOOKER STUDIO DASHBOARD J DASHBOARD K DASHBOARD L

7 Bank campaigns, from many platforms to one view

PROBLEM — A bank ran search, display, social and video campaigns. Each ad platform reported in its own format, and the team needed one view of spend, clicks and conversions, down to each step of the application form.

DATA — Ad platform exports and website conversion tags, such as brochure downloads, lead forms and applications, combined in Datorama and BigQuery.

WHAT I BUILT — Three Looker Studio dashboards: a paid media overview of traffic, spend and conversions; a search report by campaign, keyword group and product; and a search and display report that follows the application form step by step, with cost per step and written insights.

OUTCOME — **Every channel's spend and results in one place, on the same definitions.** The team can see where applicants drop out of the form, what each step costs, and which sites and keywords cost the most for what they bring.

SOCIAL MEDIA ANALYSIS · FACEBOOK GROUPS DASHBOARD D

8 Daily tracking of liver-health Facebook groups

PROBLEM	The liver-health program ran Facebook groups where members and admins post every day. The team needed a steady record of what was posted, by whom, and how members responded.
DATA	Daily posts from members (user-generated content) and admins, with their comments and reactions.
WHAT I BUILT	A daily post tracker that separates member posts from admin posts and counts the comments and replies each one gets. It shows posting by day and weekday, and which admin posts start the most conversation.
OUTCOME	**The team can see how active the groups are each day and which admin posts get members talking.** Admin posts can be planned around what works.

SYNTHESIZED ANALYSIS · AI TEXT CLASSIFICATION (LLM) SAMPLE DECK

9 How treatment-ready is the community?

PROBLEM	Beyond counts, the team needed to understand the community: what members care about, who leads the conversation, and how ready members are to act on treatment.
DATA	Thousands of member posts and comments from the daily tracker.
WHAT I BUILT	I used our work AI tool, a large language model, as a text classifier. One fixed prompt template with set categories and a JSON answer meant every row was coded the same way: a theme such as community support, lifestyle or treatment questions, a behaviour such as asking for advice, and a treatment-readiness stage. I found the top voices, synthesized the results, and presented them in a quarterly report deck.
OUTCOME	**A clear picture of what members need and how ready they are to talk about treatment.** Each quarterly report gave the team themes to build content around and the share of members ready to ask a doctor.

Dashboard recreations

Twelve dashboards in three industries, rebuilt for this page. Each one follows the sections, filters and metrics of my original in Looker Studio, and each has its own colours. The filter bars show the filters the originals had. Every number is sample data, made up for this page.

Pharma · liver-health awareness website and Facebook groups

(A) Visitor journey

Jan – Mar · vs Oct – Dec · SAMPLE DATA

ACCOUNT **Pharmaceutical company** PRODUCT **Liver-health awareness website**

FILTERS Date range

Sessions	Engaged sessions	Engagement rate
48,210	**26,050**	**54.0%**
▲ 12.4%	▲ 9.8%	▲ 1.3 pts

JOURNEY STAGES AND THEIR KEY ACTIONS

Interest			Intent		Action
Exploring	Returning	Stated interest	Preparing	Stated intent	Acting
Key actions	Sign-up and webinar submits	Concern poll submits	Guide download, advocacy tool	Testing poll submits	Specialist poll, finder searches, brand links
9.6%	**4.1%**	**1.2%**	**3.9%**	**0.8%**	**0.6%**
▲ 0.8 pts	▲ 0.3 pts	▼ 0.1 pts	▲ 0.6 pts	▲ 0.1 pts	▼ 0.2 pts
4,630 sessions 12,410 events	1,980 sessions 2,140 events	580 sessions 610 events	1,880 sessions 2,960 events	390 sessions 405 events	290 sessions 330 events

Each stage shows % of sessions engaged (sessions with the stage's key actions ÷ all sessions), then sessions engaged and total events.

(B) Site performance

December · vs November · SAMPLE DATA

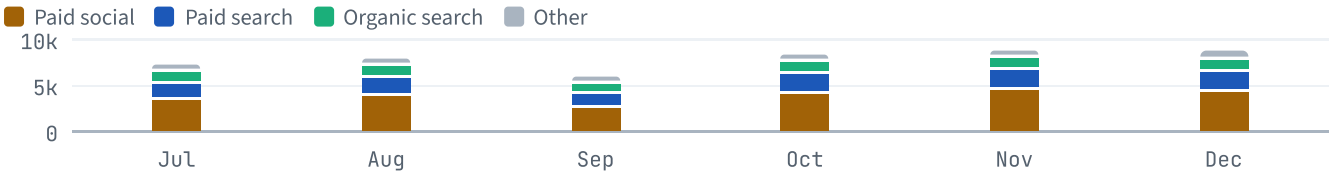
ACCOUNT **Pharmaceutical company** PRODUCT **Liver-health awareness website**

FILTERS Month

TOPLINE SITE PERFORMANCE

Sessions	Total users	Avg. session	Engagement rate	Interest goal rate	Intent goal rate
8,940	**7,410**	**03:10**	**50.7%**	**1.80%**	**0.78%**
▼ 1.7%	▼ 1.2%	▲ 8.2%	▲ 1.7 pts	▲ 0.3 pts	▲ 0.1 pts

Total site sessions by channel

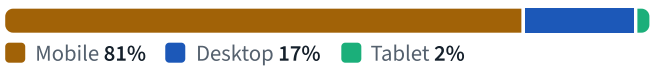


Total sessions by channel



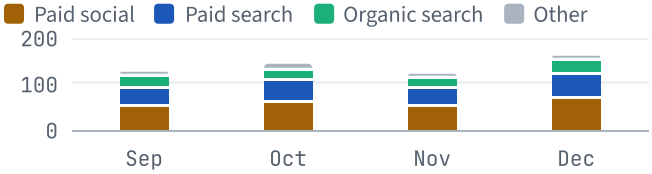
Paid social **56%** · Paid search **24%** · Organic search **12%** · Other **8%**

Total sessions by device

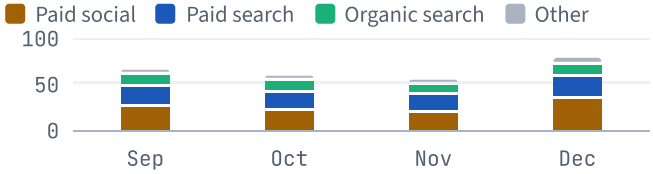


Mobile **81%** · Desktop **17%** · Tablet **2%**

Interest-segment sessions by channel



Intent-segment sessions by channel



Interest and intent goal rates = sessions that answered the interest or intent poll ÷ all sessions.

Goal conversions and key actions

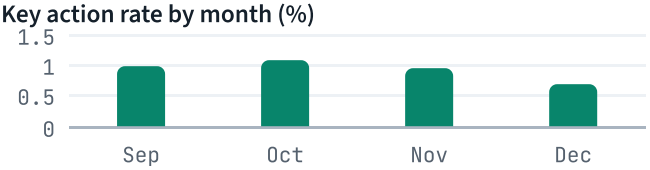
Sep – Dec | SAMPLE DATA

ACCOUNT **Pharmaceutical company** PRODUCT **Liver-health awareness website**

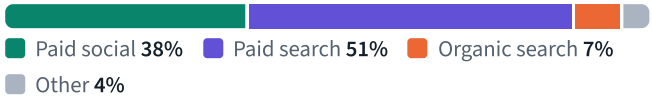
INTEREST SEGMENT

Key actions	Key action rate	Conversion rate
465	**0.84%**	**0.31%**
▲ 2.0%	▼ 0.15 pts	▲ 0.02 pts

Key action rate by month (%)



Key actions by channel

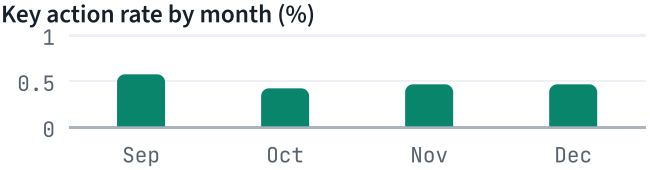


Paid social **38%** Paid search **51%** Organic search **7%** Other **4%**

INTENT SEGMENT

Key actions	Key action rate	Conversion rate
380	**0.69%**	**0.16%**
▲ 3.8%	▲ 0.04 pts	▲ 0.01 pts

Key action rate by month (%)



Key actions by channel

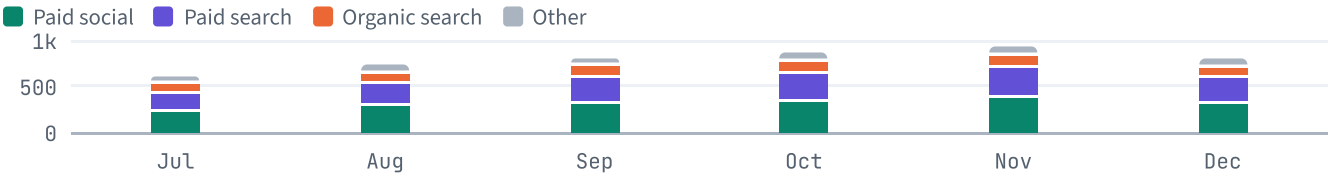


Paid social **41%** Paid search **48%** Organic search **8%** Other **3%**

KEY ACTIONS

Site sessions	Key actions	Key action rate
48,210	**4,812**	**9.98%**
▲ 12.4%	▲ 18.6%	▲ 0.5 pts

Key actions by month



Key actions by event

EVENT NAME	SESSIONS	CHANGE
form_start	1,820	▲ 21.4%
slider_progress	1,460	▲ 18.0%
form_submit	1,310	▲ 15.2%
poll_concern	960	▲ 9.7%
webinar_complete	690	▲ 27.3%
poll_submit	520	▼ 4.1%
webinar_register	455	▲ 12.6%
test_request_guide	400	▲ 8.9%

Facebook groups: daily posts

Last 8 weeks · vs 8 weeks before

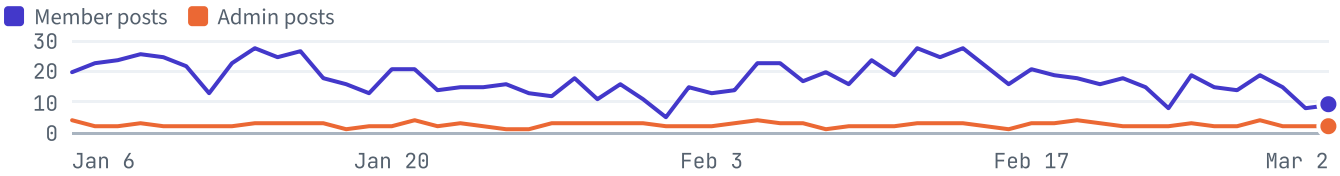
SAMPLE DATA

ACCOUNT **Pharmaceutical company** PRODUCT **Liver-health Facebook groups**

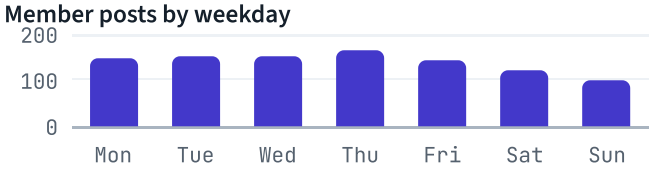
FILTERS Group ▾ Posted by ▾ Format ▾ Date range ▾

Member posts	Admin posts	Comments	Active members
1,008	138	7,560	1,940
▲ 12.0%	▲ 4.5%	▲ 15.3%	▲ 8.7%

Posts by day



Member posts by weekday



Admin posts that started the most conversation

ADMIN POST	COMMENTS	MEMBERS REPLYING
Ask a specialist (live)	412	268
Myth vs fact	236	171
Member story of the week	198	142
Weekly check-in poll	187	160
Meal idea	121	94

Infant nutrition · formula milk brand

Intercept survey

Last 8 months · SAMPLE DATA

ACCOUNT **Infant-nutrition brand** PRODUCT **Formula milk, brand website**

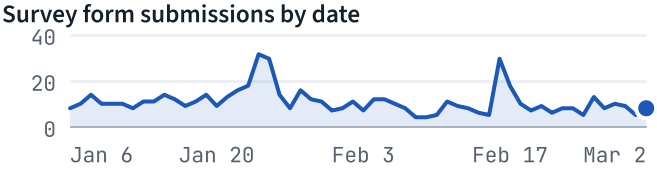
FILTERS Device ▾ | UTM source ▾ | UTM medium ▾ | Page type ▾ | Date range ▾

Pop-ups	Form submissions	Submission rate*	Form impression rate
26,150	**612**	**2.34%**	**6.4%**
▲ 6.2%	▲ 14.0%	▲ 0.16 pts	▲ 0.3 pts

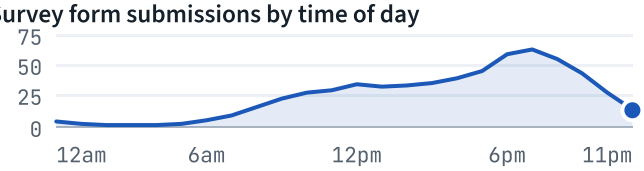
**Form submissions were tracked from a later date than pop-ups, so the submission rate counts from that date on.*

SUBMISSIONS OVER TIME

Survey form submissions by date



Survey form submissions by time of day



SUBMISSIONS BY UTM SOURCE / MEDIUM

Submissions by UTM source + medium



DEVICE BREAKDOWN

Pop-up sessions by device



Mobile **70%** Desktop **29%** Tablet **1%**

Survey submissions by device



Mobile **74%** Desktop **26%**

PRODUCT RANGE PAGES VS OTHER PAGES, SIDE BY SIDE

Pop-ups by page type

Other pages	22,290
Product range	3,860

Submissions by page type

Other pages	508
Product range	104

Submission rate by page type

Product range	2.7%
Other pages	2.3%

POP-UPS VS SUBMISSIONS BY PAGE

Top pages by pop-ups

/latest-promotion	8,420
/ (home)	5,860
/first-purchase-offer	2,140
/range-classic	1,720
/products	1,620
/checklist	1,450

Top pages by submissions

/ (home)	96
/latest-promotion	92
/range-gold	31
/range-classic	26
/range-kids	22
/products	15

Top pages by submission rate

/member-update	12.0%
/trial-offer	10.7%
/preparation-guide	9.1%
/range-premium	8.6%
/range-kids	4.7%
/range-gold	3.1%

Product range page performance

PAGE PATH	POP-UPS	SUBMISSIONS	RATE
/range-classic	1,720	26	1.5%
/range-gold	1,010	31	3.1%
/range-organic	520	13	2.5%
/range-kids	470	22	4.7%
/range-premium	140	12	8.6%
All range pages	**3,860**	**104**	**2.7%**

Other page performance

PAGE PATH	POP-UPS	SUBMISSIONS	RATE
/latest-promotion	8,420	92	1.1%
/ (home)	5,860	96	1.6%
/first-purchase-offer	2,140	–	–
/products	1,620	15	0.9%
/checklist	1,450	6	0.4%
All other pages	**22,290**	**508**	**2.3%**

CTA performance

Last 6 weeks · vs 6 weeks before

SAMPLE DATA

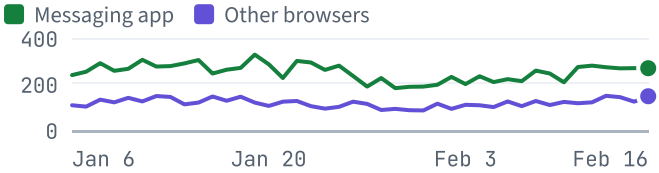
ACCOUNT **Infant-nutrition brand** PRODUCT **Formula milk, two campaign sites**

FILTERS Site ▾ | CTA ▾ | Device ▾ | Source / medium ▾ | Product name ▾ | Product ID ▾ | Date range ▾

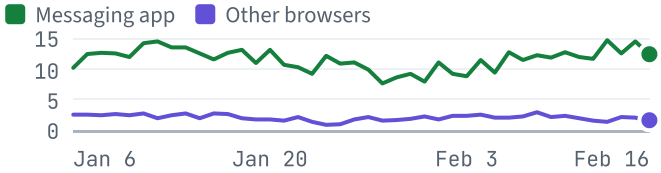
Total sessions	Total users	Messaging-app sessions	Other sessions	CTA click-through
15,920	**11,040**	**10,840**	**5,080**	**8.5%**
▼ 4.1%	▼ 3.6%	▼ 2.0%	▼ 8.3%	▲ 0.6 pts

CTA click-through = sessions with at least one click on any of the three CTAs ÷ all sessions.

Sessions over time



CTA click-through over time (%)



CTA clicks by CTA

3-minute quiz	1,046
Product report	251
Member club	56

CTA sessions by device



Mobile **90%** Desktop **9%** Tablet **1%**

CTA sessions by product

(not set)	7,320
Product A	2,140
Product B	1,860
Product C	1,410
Product D	990
Other products	2,200

CTA performance

CTA	SESSIONS	USERS	CLICKS	CTR
3-minute quiz	1,010	980	1,046	6.57%
Product report	238	221	251	1.58%
Member club	51	49	56	0.35%

Source / medium by CTA

SOURCE / MEDIUM	CTA	CLICKS	CTR
(direct) / (none)	Quiz	912	5.73%
(not set)	Quiz	70	0.44%
(not set)	Report	48	0.30%
messaging app / referral	Quiz	31	0.19%
google / organic	Quiz	22	0.14%

Organic search

Last month · vs month before SAMPLE DATA

ACCOUNT **Infant-nutrition brand** PRODUCT **Formula milk, brand website** DATA **GA4, Search Console**

FILTERS Page group ▾ Query ▾ Date range ▾

Search clicks	Impressions	Click-through rate	Avg. position
9,840	**412,300**	**2.39%**	**8.2**
▲ 4.2%	▲ 7.9%	▼ 0.08 pts	▲ 0.4 better

SEARCH CONSOLE: QUERY GROUPS

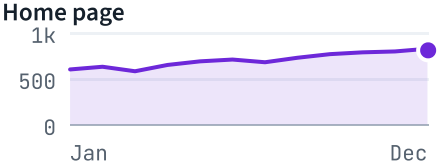
QUERY GROUP	CLICKS	CHANGE	AVG. POSITION
Overall	**9,840**	▲ 4.2%	8.2
Branded	4,210	▲ 3.1%	3.6
Products	2,180	▼ 6.5%	13.7
Branded sample	0	–	no data
Generic sample / member club	960	▲ 22.2%	3.6
Hubs	2,040	▼ 8.3%	7.1
Content hub	650	▼ 0.5%	7.2

Top queries: hubs

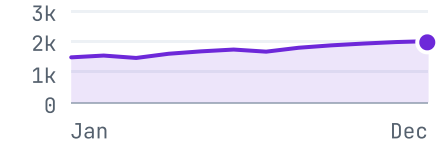
QUERY	CLICKS
growth spurt signs	48
what contractions feel like	25
baby refusing milk	22
hospital bag checklist	22
false labour	16

12-MONTH TRENDS: SEARCH CLICKS BY PAGE GROUP

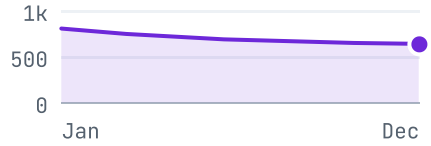
Home page



Hubs



Content hub



GA4: ORGANIC SESSIONS BY PAGE GROUP

PAGE GROUP	SESSIONS	CHANGE	BOUNCE RATE	CHANGE	AVG. SESSION
Guides	2,950	▲ 6.1%	31.9%	▼ 1.2 pts	01:49
Product pages	2,770	▲ 1.6%	21.6%	▼ 0.5 pts	02:11
Home	1,420	▼ 2.5%	15.9%	▲ 0.1 pts	01:19
Content hub	720	▼ 7.5%	30.8%	▼ 4.2 pts	02:01
Member club	210	▼ 8.3%	6.2%	▼ 2.3 pts	02:28
Other pages	7,090	▲ 2.0%	28.6%	▼ 0.4 pts	02:45
All pages	**15,160**	▲ 1.6%	**27.0%**	▼ 0.7 pts	**02:29**

Visits referred by AI assistants

LANDING PAGE	SESSIONS	BOUNCE	AVG. SESSION
/ (home)	5	60%	00:04
/range-gold	3	67%	00:00
/campaign	2	0%	00:40
/range-premium	2	50%	00:13
All AI referrals	**46**	**48%**	**02:59**

Keyword performance: one hub

QUERY	IMPR.	CLICKS	CTR	POSITION
colic in babies	613	3	0.49%	3.9
baby constipation	330	2	0.61%	7.5
baby poo colours	288	0	0%	6.3
colic remedies	272	5	1.84%	4.1
Hub total	**37,374**	**601**	**1.61%**	**6.8**

Global websites

Aug – Oct · SAMPLE DATA

ACCOUNT **Infant-nutrition brand** · PRODUCT **Formula milk, websites in seven countries**

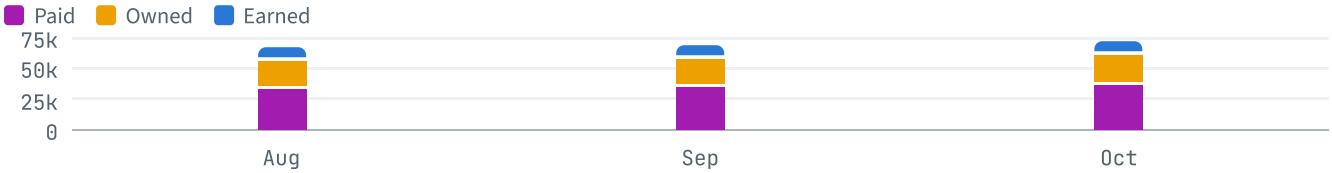
FILTERS: Country · Channel · Page type · Currency · Date range

Catch · Connect · Convert · Webshop · Continue

Users	Engaged sessions	Sticky CTA clicks	Purchase rate
214,600	**61%**	**10,110**	**1.8%**
▲ 5.2%	▲ 1.4 pts	▲ 6.4%	▲ 0.2 pts

CATCH: MONTHLY USERS BY CHANNEL BUCKET

Users by channel bucket



CONNECT: USERS, ENGAGEMENT AND SCROLL DEPTH BY MARKET

MARKET	USERS	ENGAGED	PURCHASE RATE
Market A	58,200	63%	2.1%
Market B	41,700	58%	1.6%
Market C	36,900	61%	1.9%
Market D	29,400	55%	1.2%
Market E	22,800	66%	2.4%
Market F	15,300	59%	1.4%
Market G	10,300	62%	1.7%

Users reaching each scroll depth

MARKET	25%	50%	75%	90%
Market A	78%	52%	31%	18%
Market B	74%	49%	28%	15%
Market C	81%	57%	35%	21%
Market D	69%	44%	24%	12%
Market E	83%	60%	38%	24%

CONVERT: STICKY CTA CLICKS BY CHANNEL

Primary sticky CTA clicks

Paid search	3,120
Paid social	2,560
Organic search	1,890
Direct	1,320
Referral	780
Email	440

WEBSHOP: TRAFFIC BY CHANNEL BUCKET

BUCKET	USERS	PURCHASES	RATE
Paid	119,100	2,050	1.7%
Owned	66,100	1,390	2.1%
Earned	29,400	420	1.4%
All	**214,600**	**3,860**	**1.8%**

CONTINUE: NEW VS RETURNING USERS

USER TYPE	USERS	ENGAGED SESSIONS	REVENUE PER USER
New	171,700	58%	0.42
Returning	42,900	71%	1.96

Revenue per user is shown in each market's own currency, picked with the currency filter. A glossary page in the original defines every metric.

Pack-scan site

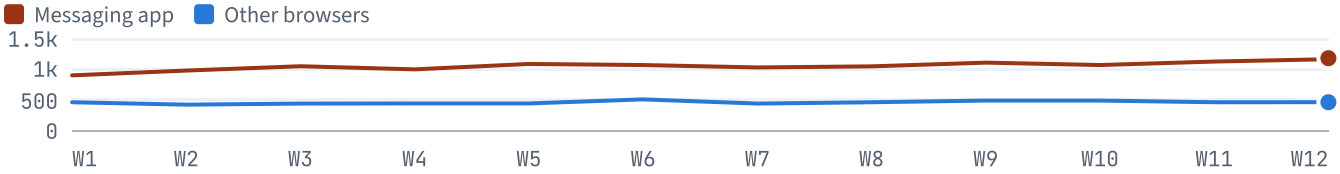
Last 12 weeks · SAMPLE DATA

ACCOUNT **Infant-nutrition brand** PRODUCT **Pack-scan site in seven countries**

FILTERS Country ▾ Product ▾ Date range ▾

Total scans	Messaging-app scans	Other scans	Scan accuracy*
18,400	**12,827**	**5,573**	**98.9%**
▲ 9.1%	▲ 11.2%	▲ 4.6%	▼ 0.3 pts

Weekly scans



Scans by market

Market A	5,980
Market B	3,870
Market C	2,940
Market D	2,310
Market E	1,640
Market F	1,020
Market G	640

Scans by city tier, one market

Tier 1	46%
Tier 2	31%
Tier 3	15%
Tier 4 and below	8%

Scan errors by type

Code not recognised	96
Already scanned	58
Expired code	34
Other	22

Product breakdown

PRODUCT	SCANS	SHARE	ERRORS
Product A, stage 3	2,130	11.6%	21
Product B, stage 2	1,860	10.1%	18
Product C, stage 1	1,740	9.5%	24
Product D, stage 3	1,290	7.0%	11
Other products	11,380	61.8%	136

**Scan accuracy = 1 − scan errors ÷ scans. 210 errors in 18,400 scans.*

Banking · digital campaigns

Search and display spend

Last 12 weeks · SAMPLE DATA

ACCOUNT **Bank** PRODUCT **Search and display campaigns** DATA **Datorama, BigQuery**

Overview | Search | Display

FILTERS Date ▾ Site ▾ Month ▾ Week ▾

PERFORMANCE SUMMARY

COST	IMPR.	CLICKS	CTR	CPC	CPM	PRODUCT PAGE	FORM START	DETAILS	SUBMITTED	CVR
$215,800	6.52M	41,730	0.64%	$5.17	$33.10	23,280	20,300	3,580	2,410	5.78%

Application form steps, and cost per step

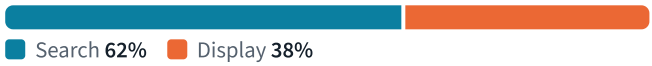


SPENDS

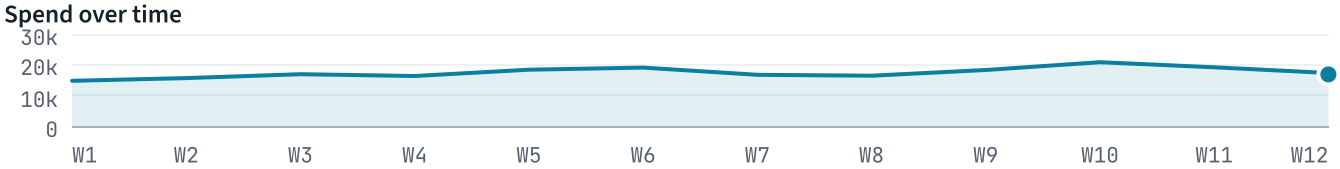
Spend per site (display)



Spend per channel



Spend over time



CPC	Cost per form start	Cost per details step	Cost per submitted form
$5.17	**$10.63**	**$60.28**	**$89.54**
▼ 4.4%	▼ 1.3%	▲ 2.2%	▼ 4.8%

What the numbers say

- The video site was the most expensive display site, with a total cost of $31,200, 38% of display spend.
- 12% of started forms were submitted, at $89.54 per submitted form.
- Search took 62% of spend.

Search campaign report

Last 30 days | SAMPLE DATA

ACCOUNT **Bank** PRODUCT **Search campaigns** DATA **Datorama, BigQuery**

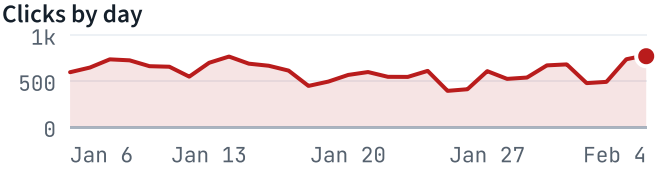
Overview | Monthly breakdown | Weekly breakdown | Product breakdown | Ads breakdown

FILTERS Search engine | Campaign cluster | Product | Date range

IMPRESSIONS AND CLICKS

Impressions	Clicks	CTR	CPC	Cost
1,151,900	**18,250**	**1.58%**	**$4.91**	**$89,590**
▲ 5.8%	▲ 3.4%	▼ 0.04 pts	▲ 2.1%	▲ 5.6%

Clicks by day



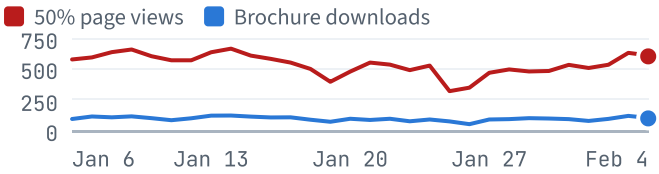
Product ranking by lead forms



CONVERSION TAGS

50% page views	Brochure downloads	Lead forms	Applications	Cost per download
16,530	**2,920**	**55**	**48**	**$30.68**
▲ 7.1%	▲ 0.7%	▲ 5.8%	▲ 4.3%	▲ 4.9%

50% page views and brochure downloads



Top 5 keyword groups by clicks



Overall and individual campaign performance

CAMPAIGN	ENGINE	IMPR.	CLICKS	CTR	CPC	COST	DOWNLOADS	LEADS
Brand search	Google	149,800	9,140	6.1%	$1.92	$17,550	1,390	24
Product search	Google	274,200	5,210	1.9%	$6.10	$31,780	910	19
Generic search	Google	292,200	2,630	0.9%	$8.40	$22,090	420	8
Competitor terms	Bing	435,700	1,270	0.3%	$14.31	$18,170	200	4
All campaigns		**1,151,900**	**18,250**	**1.58%**	**$4.91**	**$89,590**	**2,920**	**55**

Paid media overview

Jul – Dec · vs Jan – Jun SAMPLE DATA

ACCOUNT **Bank** PRODUCT **Website and paid media** DATA **Datorama, BigQuery**

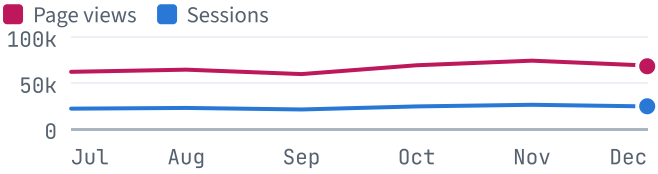
Overview | Channel

FILTERS Campaign ▾ Channel ▾ Division ▾ Month ▾

TRAFFIC

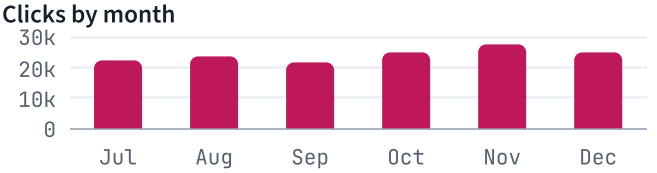
Page views	Sessions	Bounce rate	Avg. session
402,800	140,000	60.4%	04:12
▲ 7.5%	▲ 7.3%	▲ 6.6 pts	▼ 8.1%

Sessions and page views



Impressions	Clicks	CTR	CPC
11.9M	147,526	1.24%	$1.27
▲ 6.3%	▲ 6.7%	▲ 0.01 pts	▼ 0.8%

Clicks by month



SPENDS AND CONVERSION

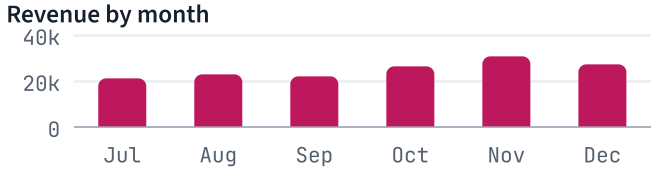
Revenue	Total cost	Adds to cart
$153,070	$187,210	956
▲ 9.2%	▲ 7.3%	▲ 8.6%

Add-to-cart rate	Purchases	CPA
0.65%	474	$394.96
▲ 0.01 pts	▲ 9.0%	▼ 1.6%

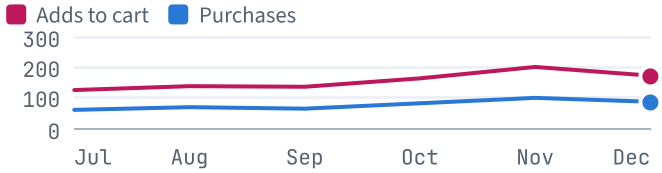
Cost per platform



Revenue by month



Adds to cart and purchases



Sample deck: quarterly community report

A recreation of the quarterly report from case study 9, the synthesized analysis of liver-health Facebook groups. Six slides, with sample data and no member names.

QUARTERLY REPORT · Q3 · SAMPLE DATA

What members said in the liver-health Facebook groups

A synthesized analysis of 8,568 member posts and comments

Summary

25% of posts and comments are about community support, the top theme.

34% ask for advice, more than any other behaviour.

24 top voices lead most conversations.

16% of active members are ready to ask a doctor, up 3 points.

Method: AI text classification

Thousands of rows were classified with our work AI tool, a large language model. Every row got the same prompt and answered in the same JSON format, so the coding stayed uniform.

```
Read this post or comment from a
liver-health support group.
Return JSON with:
- theme: community support | lifestyle |
  diagnosis and tests | treatment
  questions | symptoms | other
- behaviour: asking for advice | sharing
  an experience | offering support |
  sharing resources | venting | other
- readiness: just joined | learning |
  considering treatment | ready to
  ask a doctor
```

Rewritten example, not the original prompt.

What members talk about, and how

Themes in posts and comments



Behaviours, share of all



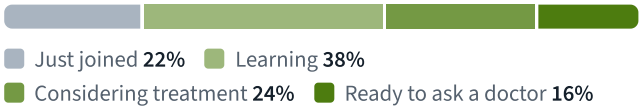
Top voices

MEMBER	POSTS	COMMENTS	MAIN THEME
Member A	38	412	Community support
Member B	31	356	Lifestyle
Member C	27	298	Treatment questions
Member D	22	264	Community support

Most of them share their own experience and answer others' questions.

Is the community treatment-ready?

Treatment readiness of active members



Next quarter

- Plan admin posts around community support and lifestyle.
- Invite top voices to share their experience.
- Answer treatment questions in a live specialist session.

GA4 event tracking

A developer sets up the tags in Google Tag Manager. My part is deciding what to track and making it reportable: I keep the list of event and click names on each website, register custom parameters in GA4, and build the dashboards on them.

1 · MAP THE EVENTS

Every event and click name, and what it means

Event names · Click names · Key actions · Conversions

2 · CUSTOM PARAMETERS

Register the details as custom dimensions

Survey answers · CTA names · Page types · File names

3 · REPORT

Build the dashboard on events and their parameters

Looker Studio · Answers by option · Clicks by CTA · Rates by stage

Tracking plan

SAMPLE DATA

Events, their parameters, and where they're reported

EVENT	FIRES WHEN	PARAMETERS	REPORTED AS
`poll_submit`	A visitor answers a survey question	`poll_question`, `poll_answer`	Answers by option
`cta_click`	A visitor clicks a call to action	`cta_name`, `page_type`	Clicks and click-through by CTA
`file_download`	A visitor downloads a guide	`file_name`	Key actions by journey stage
`video_progress`	A video reaches 25, 50 or 75%	`video_title`, `video_percent`	Viewing depth by video
`finder_search`	A visitor uses the location finder	`search_type`	Searches by type
`form_submit`	A visitor signs up	`form_name`	Sign-ups by form

Custom dimensions in GA4

Admin › Data display › Custom definitions

SAMPLE DATA

Custom dimensions registered from event parameters

DIMENSION NAME	SCOPE	EVENT PARAMETER	USED FOR
Poll question	Event	`poll_question`	Which question was answered
Poll answer	Event	`poll_answer`	Survey answers by option
CTA name	Event	`cta_name`	Clicks by call to action
Page type	Event	`page_type`	Product pages vs other pages
File name	Event	`file_name`	Downloads by guide
Video title	Event	`video_title`	Viewing depth by video

How I register one

1. Check the parameter arrives with its event in GA4.
2. Open Admin › Data display › Custom definitions, then Create custom dimension.
3. Give it a readable name, set the scope to Event, and enter the parameter name exactly as it's sent, such as `poll_answer`.
4. Save, then use the new dimension in reports and Looker Studio.

GA4 fills a custom dimension only from the day it's created, so a parameter needs registering before its data is needed.

Survey answers: "How useful was this page?"



The `poll_answer` dimension in use: answers by option.

SQL and code

Short generic examples of the patterns behind the case studies, written for this page. Client code isn't copied, and table names are made up.

PYTHON · AWS LAMBDA

① Load each report as it arrives

A file landing in S3 starts the function. It finds the report type from the file name, loads the rows, then moves the file out of the inbox.

```
import io, os, boto3
import pandas as pd
from urllib.parse import unquote_plus
from sqlalchemy import create_engine

s3 = boto3.client("s3")
db = create_engine(os.environ["DB_URL"])
TABLES = {"business-report": "sales",
          "sp-ads": "ads_sp",
          "inventory": "inventory"}

def handler(event, context):
    rec = event["Records"][0]["s3"]
    bucket = rec["bucket"]["name"]
    key = unquote_plus(rec["object"]["key"])
    table = next(TABLES[k] for k in TABLES
                 if k in key)
    obj = s3.get_object(Bucket=bucket,
                        Key=key)
    df = pd.read_csv(io.BytesIO(
        obj["Body"].read()))
    df.to_sql(table, db, schema="raw",
              if_exists="append",
              index=False)
    # Move the file: raw/ to done/
    s3.copy_object(Bucket=bucket,
        Key=key.replace("raw/", "done/"),
        CopySource=f"{bucket}/{key}")
    s3.delete_object(Bucket=bucket, Key=key)
```

SQL · POSTGRESQL

② Copy only new rows, then move the bookmark

The sync reads where it stopped last time, loads only rows added since, and moves its bookmark no further than the rows it just loaded.

```
-- Runs in one transaction
create temp table new_rows
  on commit drop as
select s.*
from staging.search_terms s
where s.loaded_at > (
  select last_loaded_at
  from sync_state
  where source = 'search_terms');

insert into mart.search_terms_weekly
  (week_start, product_id, term,
   impressions, clicks, purchases)
select week_start, product_id, term,
       impressions, clicks, purchases
from new_rows
where week_end - week_start <= 6;

update sync_state
set last_loaded_at = coalesce(
  (select max(loaded_at) from new_rows),
  last_loaded_at)
where source = 'search_terms';
```

SQL · BIGQUERY

③ One daily table across ad platforms

Each platform names its columns differently. One model lines them up, so every dashboard reads the same table.

```
-- One row per day, channel, campaign
create or replace table mart.daily_ads
partition by date as
select date, 'Search' as channel,
       campaign, cost, clicks,
       conversions
from raw.search_ads
union all
select date, 'Social', campaign_name,
       spend, link_clicks, purchases
from raw.social_ads;

-- Cost per conversion by week
select date_trunc(date, week) as week,
       channel,
       safe_divide(sum(cost),
                   sum(conversions)) as cpa
from mart.daily_ads
group by week, channel
order by week, channel;
```

SQL · SNOWFLAKE

④ Mirror shared data on a schedule, then check it

A data provider shares Amazon seller data with us in Snowflake. A scheduled task copies it into our own tables, and a count checks the copy matches its source. The PostgreSQL sync then moves new rows to AWS.

```
-- Refresh our mirror of the shared table
create or replace task refresh_daily_mirror
  warehouse = reporting_wh
  schedule = 'USING CRON 0 6 * * * UTC'
as
  create or replace table
    mirror.daily_sales as
  select * from shared.amazon.daily_sales;

-- Tasks start paused
alter task refresh_daily_mirror resume;

-- The mirror must match its source
select 'mirror' as tbl, count(*) as n
from mirror.daily_sales
union all
select 'source', count(*)
from shared.amazon.daily_sales;
```

How I work

1. **Start from the question**
   I turn the business ask into the metrics, filters and time periods the report needs.
2. **Check the tracking**
   I list the events, tags and conversions each metric needs, and check they fire in GA4 before I build on them.
3. **Land the raw data**
   Scheduled jobs pull each source into S3 or the warehouse and keep the original files.
4. **Model it with SQL**
   I clean, join and aggregate into tables at the grain the report needs, such as one row per product per day.
5. **Test before it ships**
   I run new jobs on a test copy first, then check that required files are present, nothing is counted twice, and row counts and totals match the source.
6. **Design the dashboard**
   Summary numbers first, then trends, then detail tables, with every metric defined on the page.
7. **Report what changed**
   Monthly and quarterly decks: what moved, why, and what to try next.

Tools and skills

SQL and modelling

BigQuery · Snowflake · PostgreSQL · Joins and window functions · Data models · Incremental loads · Data-quality checks

Pipelines and cloud

AWS S3 · Lambda · EventBridge · EC2 · RDS · CloudWatch · SSH tunnels · Snowflake tasks · Python · pandas · boto3 · Datorama · Google Apps Script

Dashboards

Looker Studio · Power BI · Lovable · KPI design · Metric glossaries

Web and marketing data

GA4 · GTM · Search Console · Events and conversions · Custom dimensions · UTM tracking · Google Ads · Bing Ads · Meta and TikTok ads · YouTube · Amazon seller reports

Analysis and reporting

Funnels and journeys · Campaign performance · SEO · Social media and sentiment · Executive summaries · Insight decks · Community analysis · AI text classification (LLM) · Prompt design

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Client, brand and colleague names and all business figures are left out. The dashboards are recreations with sample data, and the code is rewritten, not copied.